

DermoHairNet

U-Net Hair Artifact Segmentation and Counting for Dermoscopy

Original project title: Automated Hair Count in Dermoscopy Using Image Processing

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Project visual summary: dermoscopic image, binary hair mask and overlay used for qualitative inspection.

U-Net CNN segmentation model	Label Studio manual annotation	OpenCV post-processing	ISIC image source context
This report presents the project as a technical research artifact for medical image preprocessing. The system is designed to detect and count hair artifacts that can obstruct dermoscopic lesion analysis. It is not a clinical diagnostic system.			

1. Abstract

Hair artifacts in dermoscopic photographs can obscure important lesion information such as colour gradients, texture patterns and lesion boundaries. This project developed an automated system to detect, segment and count hair in dermoscopic images of melanoma-affected skin. The method combines a U-Net convolutional neural network for pixel-level segmentation with OpenCV-based contour analysis for hair counting.

Training was performed on manually annotated images drawn from the ISIC dermoscopy dataset context. The research portfolio reports 200 annotated images from approximately 44,000 available dermoscopic samples, with approximately 98 percent test accuracy and approximately 97.5 percent validation accuracy. The resulting pipeline classifies images by hair count into 20 categories, enabling downstream image-analysis workflows to select cleaner dermoscopic inputs.

200 manually annotated images	about 44,000 source dataset context	about 98% reported test accuracy	about 97.5% reported validation accuracy
512 x 512 input image size	2 segmentation classes	20 hair-count categories	100 training epochs

Research objectives

- Automatically identify and segment hair pixels in dermoscopic images using deep learning.
- Generate binary masks that separate hair artifacts from the skin and lesion background.
- Count distinct hair structures using image-processing methods after segmentation.
- Classify images into hair-count categories from clean images to images with 19 or more hairs.
- Support later dermoscopy preprocessing workflows where hair obstruction must be reduced or filtered.

2. Background and Motivation

Computer-aided dermoscopy analysis depends strongly on image quality. Hair is a difficult artifact because it may appear as thin, curved, overlapping and high-contrast structures crossing the lesion. These structures can hide visual patterns that are relevant for image interpretation and can also introduce non-lesion signals into automated models.

Earlier hair handling approaches in dermoscopy commonly relied on classical image-processing methods and hand-crafted features. The motivation of this project was to investigate a deep learning approach that learns hair structures directly from annotated masks. U-Net was selected because semantic segmentation requires pixel-level localization rather than simple image-level classification.

Project scope

The scope of the work is medical image preprocessing. The project does not attempt to diagnose melanoma directly. Instead, it focuses on detecting and counting hair artifacts so that dermoscopic images can be filtered, cleaned or prioritized before downstream lesion-analysis steps.



Figure 4: Image of melanoma affected skin sample 1.



Figure 5: Annotated image of affected skin sample 1

Original dermoscopic image and its manually annotated hair mask. White pixels represent annotated hair strands and black pixels represent background.

3. Dataset and Annotation Workflow

The image source was the International Skin Imaging Collaboration archive, which contains dermoscopic images collected from medical and research institutions. The project report and research portfolio describe approximately 44,000 available dermoscopic images in the original dataset context. A subset of 200 images was manually annotated for supervised training.

Label Studio annotation

The annotation workflow used Label Studio. An early polygon-based annotation approach was found to be inefficient for this task because hair strands are thin, curved and often overlapping. The final workflow used the brush tool, which allowed more practical pixel-level marking of hair regions.

more features. This allows the labeling stage of automation to be less complex.

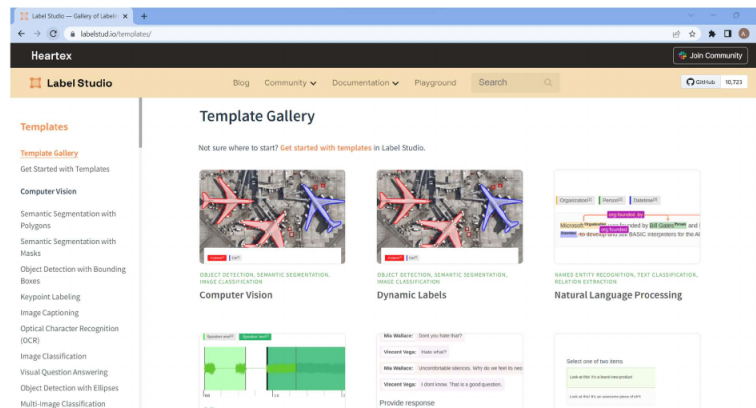


Figure 2: Label Studio Interface.

Reason for using Label Studio is that it is flexible and configurable, especially the configurable

Label Studio annotation interface from the original project report. The brush-based workflow was used for pixel-level hair segmentation masks.

Mask preparation rules

- Hair pixels were manually marked and exported as PNG segmentation masks.
- Background pixels were treated as class 0.
- Mask values from 1 to 255 were normalized to class 1 for hair.
- Images and masks were resized to 512 x 512 pixels before training.
- Masks were one-hot encoded into two channels: background and hair.
- Overlapping and partially visible hair strands were annotated so that the model would not learn them as background.

4. Preprocessing and U-Net Model

The preprocessing stage converts the annotated image dataset into arrays suitable for semantic segmentation training. Original images were converted to grayscale, resized to a fixed shape and normalized. Mask images were converted into binary labels and then categorical arrays for a two-class segmentation task.

Preprocessing pipeline

Step	Operation	Purpose
Image loading	Read dermoscopy image in grayscale	Use a single channel focused on structure and contrast.
Resizing	Resize to 512 x 512 pixels	Standardize the tensor shape for U-Net input.
Normalization	Scale image intensity values	Improve numerical stability during training.
Mask thresholding	Convert all non-zero mask pixels to 1	Create binary labels for hair versus background.
Categorical encoding	Encode masks into 512 x 512 x 2 arrays	Match the two-class softmax output of the model.
Train-validation split	80 percent training and 20 percent validation in the final project description	Evaluate model generalization on held-out images.

U-Net architecture

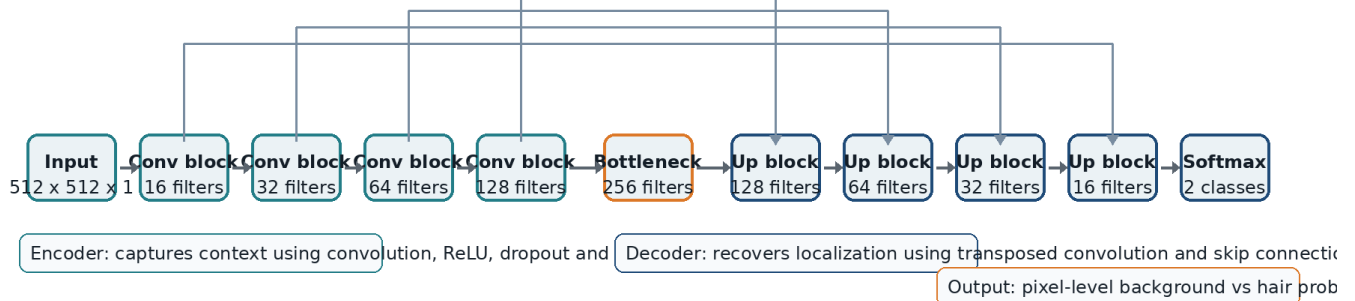
The model uses a U-Net encoder-decoder architecture. The encoder path applies convolutional blocks and max pooling to capture context, while the decoder path uses transposed convolutions and skip connections to recover spatial detail. This structure is well suited to hair segmentation because fine hair boundaries must be localized at pixel level.

Component	Implementation in the uploaded model file	Function
Input	512 x 512 x 1 grayscale image	Receives normalized dermoscopy image.
Encoder	Conv2D blocks with 16, 32, 64 and 128 filters	Learns local edges and broader texture context.
Bottleneck	Conv2D block with 256 filters and dropout	Captures high-level representation.
Decoder	Conv2DTranspose blocks with skip connections	Restores image resolution and localization detail.
Output	1 x 1 Conv2D with softmax over 2 classes	Predicts background versus hair probability per pixel.

5. End-to-End Processing Pipeline

The system is organized as a sequence of steps from annotation to model prediction and image classification. The U-Net produces a segmentation mask, and OpenCV converts that mask into a hair count. This separates the learning-based segmentation problem from the rule-based counting problem.

Simplified U-Net Architecture Implemented in segmentation.py



Simplified model and processing structure based on the uploaded U-Net implementation.

Operational workflow

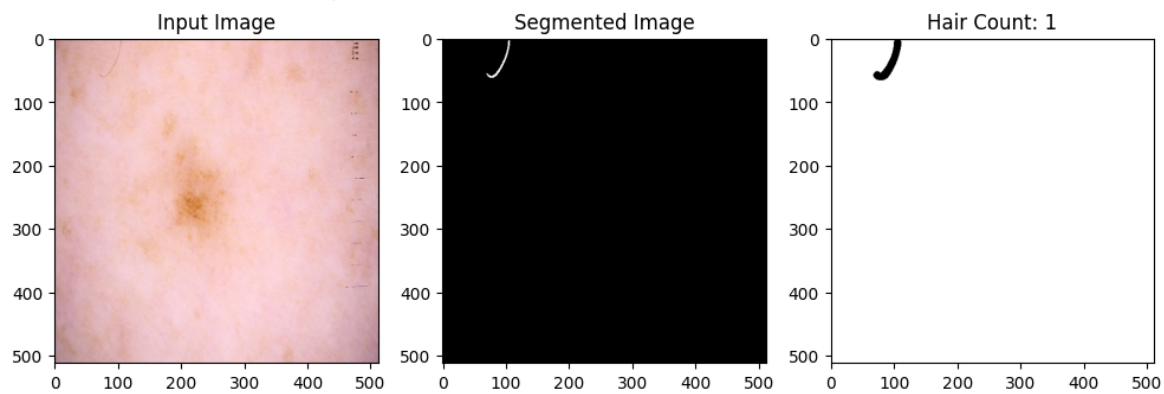
Stage	Input	Output
Annotation	Original dermoscopy images	PNG hair masks created in Label Studio
Preprocessing	Image and mask files	512 x 512 normalized image arrays and binary mask arrays
Training	Prepared image-mask pairs	Saved U-Net model file
Inference	New dermoscopy image	Predicted binary hair mask
Post-processing	Predicted mask	Contour image and hair count
Classification	Hair count value	Class label from 0 hairs to 19 or more hairs

6. OpenCV Post-Processing and Hair Count Classification

After segmentation, the predicted binary mask is processed with OpenCV. The goal is to transform the pixel-level hair region into countable objects. In the project report, the number of valid contours is used as the estimated number of hairs.

Hair counting logic

Step	Description
Binary mask generation	The U-Net output is converted into a binary hair mask.
Binary inversion	Foreground and background contrast is adjusted for contour detection.
Canny edge detection	Edges of predicted hair regions are extracted.
Contour extraction	Contours are detected from the edge map.
Contour filtering	Noise contours can be removed using an area threshold.
Counting	The number of valid contours becomes the estimated hair count.
Class assignment	The image is assigned to one of 20 classes: class 1 for zero hairs and class 20 for 19 or more hairs.



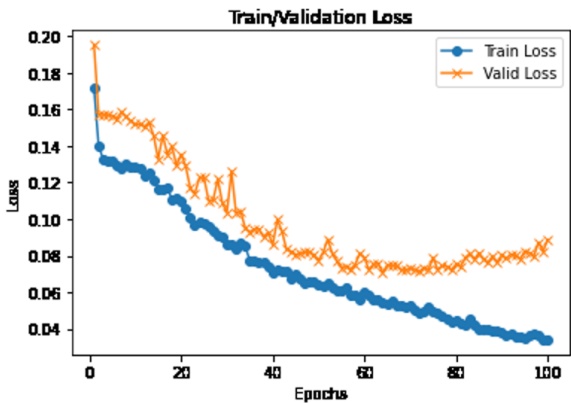
Example inference panel from the archived notebook: input image, segmented image and contour-based hair count output.

7. Training Results and Model Curves

The U-Net model was trained for 100 epochs. The reported results show strong performance for the segmentation task. The research portfolio reports approximately 98 percent test accuracy and approximately 97.5 percent validation accuracy. The archived training curves show decreasing loss and stable high validation accuracy by the final epochs.

about 98.5% training accuracy near epoch 100	about 97.5% validation accuracy	about 0.04 training loss near final epochs	100 epochs
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Curve	Observed pattern	Interpretation
Training loss	Decreased from roughly 0.17 to roughly 0.04	The model continued learning segmentation structure across training.
Validation loss	Dropped early and then stabilized with moderate fluctuation	Validation behavior suggests useful generalization, although more metrics would improve evaluation.
Training accuracy	Increased toward about 98.5 percent	The model fitted the annotated training data well.
Validation accuracy	Stabilized around the high 97 percent range	Held-out predictions remained strong for the validation split.



Loss curve



Accuracy curve

Training and validation loss and accuracy curves extracted from the archived notebook.

8. Qualitative Results

Visual validation is important because hair segmentation quality cannot be fully judged by a single accuracy score. The model must identify thin and curved hair strands while avoiding excessive false positives in the skin and lesion background.



Figure 6: Image of melanoma affected skin sample 2.



Figure 7: Annotated image of skin sample 2.

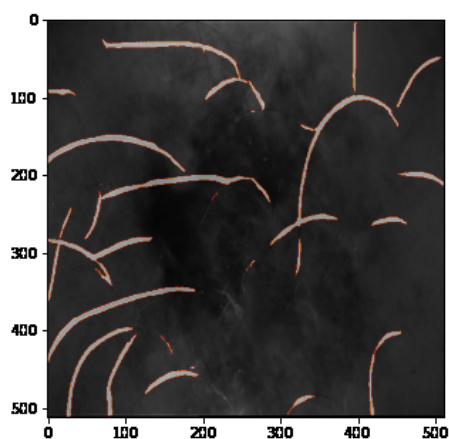


Figure 8: Image of melanoma affected skin sample 3.



Figure 9: Annotated image of skin sample 3.

Annotated examples from the original project report showing different hair density, shape and contrast.



Notebook prediction example showing detected hair structures over a dermoscopic sample.

9. Held-Out Validation Examples

The original project report included validation examples comparing raw dermoscopy images with predicted hair segmentation outputs. These examples show that the U-Net captured individual hair strands across images with different hair density and thickness.

After training the data set, the following images (20% of the data set including the following samples) were tested against the model for validation.

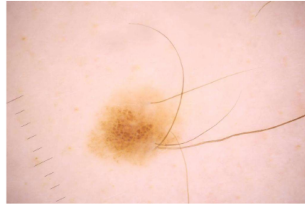


Figure 10: Test image 1.

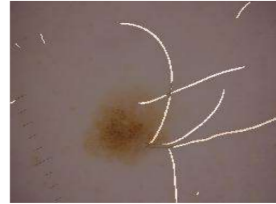


Figure 11: Prediction of hair by the model on test image 1.



Figure 12: Test image 2.



Figure 13: Prediction of hair by the model on test image 2.

Model validation examples from the original project report: dermoscopy test images and predicted hair segmentation masks.

Evaluation remarks

Accuracy is useful as an initial training metric, but future research versions should add segmentation-specific metrics such as Dice coefficient, Intersection over Union, hair-class precision and hair-class recall. These metrics would make the evaluation stronger because background pixels dominate many dermoscopic images.

10. Conclusion and Future Directions

This project demonstrated that a U-Net CNN can be used effectively for automated hair segmentation and counting in dermoscopic images. The combination of Label Studio annotation, deep learning segmentation and OpenCV contour analysis produced a complete preprocessing workflow: from image labeling to mask prediction, hair counting and class assignment.

The main contribution is not a melanoma diagnosis model, but a focused preprocessing system that can support downstream dermoscopy analysis by identifying hair obstruction. The project is valuable because it combines practical data annotation, biomedical image segmentation and interpretable post-processing in one pipeline.

Future directions

- Extend the pipeline from hair detection to hair removal using inpainting methods.
- Evaluate segmentation quality with Dice coefficient, Intersection over Union, precision and recall.
- Increase the annotated dataset to improve generalization across hair types, lighting conditions and skin tones.
- Compare U-Net with newer architectures such as attention U-Net or transfer-learning based segmentation models.
- Integrate the preprocessing module into a larger dermoscopy image analysis pipeline.

Technical skills demonstrated

Skill area	Evidence from the project
Computer vision	Pixel-level segmentation of hair artifacts in dermoscopic images.
Deep learning	U-Net CNN with encoder-decoder structure, skip connections and softmax segmentation output.
Medical image preprocessing	Focused workflow for reducing hair artifact impact before lesion analysis.
Data annotation	Brush-based Label Studio workflow for segmentation mask creation.
Python and OpenCV	Preprocessing, edge detection, contour extraction and image classification logic.
Model validation	Training curves, validation examples and qualitative inspection of predicted masks.

Prepared as a clean technical project report for public project documentation.